Saimun Habib

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RESEARCH INTERESTS

My research focuses on the foundations of safe and trustworthy AI, with a particular emphasis on neurosymbolic methods as a path toward interpretable, verifiable, and value-aligned systems. My PhD work has produced two primary contributions. First, a framework for efficient counterfactual reasoning in neuroprobabilistic logic programming via Single World Intervention Programs, and second, a characterization of the limits of value alignment via RLHF. Going forward, I am actively pursuing research on counterfactual reasoning in neurosymbolic AI, statistical and normative limits of value alignment, and neurosymbolic approaches to attention mechanisms. As a PhD Fellow at the Centre for Technomoral Futures, I am motivated by the belief that hybrid neurosymbolic architectures, offer a principled foundation for building AI systems whose behavior can be understood, audited, and reliably aligned with human values.

COMPUTATIONAL SKILLS

Python, R, C++, MATLAB, SQL, TensorFlow, Keras, Parsnip, Caret, PyTorch

EDUCATION

Ph.D., expected 2028	University of Edinburgh: School of Informatics Causality and Neurosymbolic AI
M.S., 2024	Johns Hopkins University Data Science. GPA: 3.9
A.B., 2021	Harvard University (*cum laude*) Double major in Statistics and Math (High Honors), minor in Molecular and Cellular Biology. GPA: 3.8 Thesis: Variable Importance in Composite Indices: A Sensitivity Analysis Approach for Optimized Variable Weights

PROFESSIONAL EXPERIENCE

July 2021 – Present
The MITRE Corporation
McLean, VA
Data Scientist
Operated as a research data scientist in problem domains such as social equity, computer vision, climate resilience, algorithmic fairness, causal inference, and more, to support government sponsors. Lead multiple projects and initiatives and was responsible for a $100,000 research grant. Supported multiple sponsors and regularly engaged with officials in the US Census, Federal Emergency Management Agency, Hospital of Special Surgery, Department of Homeland Security, Veterans Affairs, and Department of Treasury as an individual contributor and subject matter expert.

Jun 2020 – Jun 2021
Social and Decision Analytics Division – University of Virginia
Arlington, VA
Data Scientist
Worked as a data scientist supporting teams estimating inequities in EMS wait times in Charlottesville, VA and topic modeling the opinions of Black and White U.S. soldiers on racial integration from historic survey data. Lead a project to develop a novel statistical method optimizing weights in composite indicators/indices to create theoretically sound composites to measure complex phenomena.

Jan 2019 – Dec 2020
Dept. of Statistics – Harvard University
Cambridge, MA
Teaching Fellow
Educated and evaluated students in probability, regression modeling, descriptive statistics, and portfolio analysis. Held weekly sections to review class material in further depth to break down complex ideas into simpler components.

Jun 2019 – Aug 2019
Newton Lab
Madison, WI
Biomedical Data Science Researcher
Developed a novel Monte Carlo algorithm using C++ and R which aggregated independent gene expression studies for stronger gene set inferences.

TEACHING

Jan – May 2025
University of Edinburgh – School of Geosciences
Numeracy, Modelling and Data Management
Created, delivered, and assessed course materials on un/supervised machine learning algorithms and the ETL pipeline in R and Python.

PUBLICATIONS

Papers/White Papers/Reports/Abstracts

1. **Habib, S.**, Belle, V., He F. DeepSWIP: Quotient-WMC Counterfactuals for Neural Probabilistic Logic Programs. [Paper]. Accepted to International Conference On Neurosymbolic Learning And Reasoning. July 2026.
2. **Habib, S.**, Xiao, X., Fang, M., He F. Mission Impossible: Universal LLM Moral Alignment. [Paper]. Accepted to the Communications of ACM. July 2026.
3. **Habib, S.**, Belle, V., He F. Efficient Counterfactual Reasoning in ProbLog via Single World Intervention Program. [Paper]. Accepted to the Journal of Logic and Probability. July 2026.
4. **Habib, S.**, Munro, M., Stadlan, J., et al. Applying Causal Analysis to Equity Assessments: Racially Disparate NIH Award Rates. [Report]. Nov. 2023.
5. **Habib, S.** Spatiotemporal Causal Inference with Graphical Models: Understanding Conflict and Contraceptives in Burkina Faso. Joint Statistical Meetings Conference Proceedings. Oct. 2023.
6. **Habib, S.**, Pires, B., Benendetto, G., et al. Automated Synthetic Data Validation: Applying Noise Injection for Disclosure Avoidance. Joint Statistical Meetings Conference Proceedings. Oct. 2023.
7. Freeza, D., Orange, D., Thompson, J., **Habib, S.**, et al. Computer Vision Analysis of Rheumatoid Arthritis Synovium Reveals Lymphocytic Inflammation is Associated with Immunoglobulin Skewing in Blood. Arthritis & Rheumatology. July 2023.
8. MITRE. The Racial Wealth Gap in Washington D.C. [Report]. Dec. 2021.
9. Bohmholdt , A., Cochran F., **Habib, S.**, et al. Equity in Climate Resilience Planning and Investments. [White Paper]. Nov. 2022.

Conference Presentations

1. **Habib, S.**, Belle, V., He F. DeepSWIP: Quotient-WMC Counterfactuals for Neural Probabilistic Logic Programs. [Oral Presentation]. In the International Conference On Neurosymbolic Learning And Reasoning. Sept. 2026. *Won Best Paper Award.
2. **Habib, S.**, Xiao, X., Fang, M., He F. Mission Impossible: Universal LLM Moral Alignment. [Poster Presentation]. In Pluralistic Alignment Workshop ICML. July 2026.
3. **Habib, S.**, Belle, V., He F. Efficient Counterfactual Reasoning in ProbLog via Single World Intervention Program. [Poster Presentation]. In PhilML Workshop ICML. July 2026.
4. **Habib, S.**, Stadlan, J. Quantifying Equity: From Principles to Metrics. [Poster Presentation]. In *Harnessing the Power of Economics to Solve Whole of Nation Challenges*. McLean, VA. Sept. 2023.
5. **Habib, S.**, Pires B., Benendetto, G., et al. Automated Synthetic Data Validation: Applying Noise Injection for Disclosure Avoidance. [Oral Presentation]. In *Joint Statistical Meetings*, Toronto, Canada. Aug., 2023.
6. **Habib, S.** Towards Spatiotemporal Causal Inference with Graphical Models. [Oral Presentation]. In *Joint Statistical Meetings*, Toronto, Canada. Aug., 2023.
7. Bohmholdt , A., Cochran F., **Habib, S.**, et al. Equity in Climate Resilience Planning and Investments. [Oral Presentation]. In *National Adaptation Forum*, Baltimore, MD, USA. Oct., 2022.

MEMBERSHIPS AND ACTIVITIES

Jan 2022 – Present	American Statistical Association
Jan 2023 – Jun 2024	Volunteer at Arlington, VA Public Libraries – Conversational English Teacher
May 2023 – Present	MITRE's First Generation Professionals Committee - Co-Founder and Co-Chair
June 2026 – Present	PhD Fellow at the Centre for Technomoral Futures